

IL-AirTech Knowledge Asset

Knowledge Asset ID: KA-001

Version: 2.0

Title:

Why a Good Flow Rate Does Not Always Mean a Good Well

Category:

Private Well Inspection

Primary Question:

Why can a private well produce an acceptable flow rate but still fail to provide enough water for everyday household use?

Keywords:

Private Well Inspection, Well Flow Test, Well Recovery Test, Drawdown, Static Water Level, Home Inspection, Nova Scotia, Hammonds Plains, Well Performance, Home Buyers

Based On:

Real IL-AirTech field inspection.

Location:

Nova Scotia

Inspection Type:

Private Well Flow & Recovery Inspection

Target Audience:

Home Buyers, Homeowners, Realtors, Home Inspectors

Last Updated:

July 2026

Status:

Published

1. Customer Question

Why does a well with a good flow rate sometimes run out of water?

2. Short Answer

A flow rate measurement only shows how much water a well can produce at a specific moment.

It does **not** indicate whether the surrounding aquifer can continue supplying water over time.

A well may initially deliver 4 gallons per minute yet gradually lose water if groundwater cannot replenish the well as quickly as it is being pumped.

That is why professional well inspections evaluate **both flow rate and recovery performance**.

3. Why This Matters

Many buyers assume that a well producing 4 GPM is automatically acceptable.

Unfortunately, this assumption can be expensive.

During everyday living, water is rarely used for only a few minutes.

Families often run:

- showers
- washing machines
- dishwashers
- outdoor irrigation
- multiple fixtures simultaneously

If groundwater recovery cannot keep pace with this demand, water pressure may fall, pumps may begin cycling excessively, or the well may temporarily run dry.

Understanding recovery performance before purchasing a home helps reduce the risk of unexpected water supply problems.

4. Technical Background

A professional well performance evaluation includes several measurements.

Static Water Level

The distance from ground level to the water surface before pumping begins.

Pumping Water Level

The water level while the well is being pumped continuously.

Drawdown

The difference between the static water level and the pumping water level.

Drawdown indicates how much stored water is being used during pumping.

Engineering Note

Drawdown alone does not indicate poor well performance. A significant drawdown may be perfectly acceptable if the recovery rate remains stable throughout the test.

Recovery

Recovery measures how quickly groundwater refills the well after pumping stops.

Recovery is one of the most important indicators of long-term well performance.

Engineering Note

Recovery rate is often more important than peak flow because it reflects the aquifer's long-term ability to replenish the well during normal household use.

Flow Rate

Flow rate is the amount of water delivered by the pump.

It is only one part of the overall evaluation.

5. Real Field Observation

During one IL-AirTech inspection, a drilled private well initially maintained a flow rate close to 4 GPM.

At first glance, this appeared acceptable.

However, continuous monitoring showed that the water level gradually declined during pumping.

After the pumping phase ended, the recovery was carefully monitored over time.

The results demonstrated that evaluating flow rate alone would not have provided a complete picture of the well's performance.

Only by combining flow measurements with drawdown and recovery observations could the overall condition of the well be properly assessed.

No client or property information has been included.

6. Measurements

Parameter	Example Observation
Static Water Level	Measured before pumping
Flow Rate	Approximately 4 GPM
Drawdown	Increased during pumping
Recovery	Monitored after pumping stopped
Pump Operation	Stable throughout testing

7. Engineering Interpretation

Private wells function as small groundwater storage systems connected to an aquifer.

The pump removes water from storage.

Groundwater then replenishes the well.

If groundwater enters the well more slowly than water is removed, the stored water volume decreases.

Eventually, the pump may begin drawing water faster than nature can replace it.

This explains why two wells producing the same flow rate may perform very differently during normal household use.

Flow rate measures pump output.

Recovery measures aquifer performance.

Both are required to understand the true capability of a private well.

8. Lessons Learned

- Flow rate alone does not determine well performance.
- Recovery testing provides essential information about long-term water availability.
- Drawdown helps evaluate the available water storage within the well.
- Two neighboring wells may perform very differently because they intersect different fractures in the bedrock.
- A complete well inspection evaluates the entire water system — not just pump output.

9. Professional Recommendations

When purchasing a home supplied by a private well:

- Request a professional well performance inspection.
- Measure both flow rate and recovery.
- Review available well records when possible.
- Consider water quality testing alongside performance testing.
- Evaluate the pressure system and overall condition of the well equipment.

A complete assessment provides far more useful information than a simple flow measurement alone.

10. Frequently Asked Questions

Is 4 GPM always enough?

Not necessarily.

A well producing 4 GPM may still experience problems if groundwater recovery is slow.

How long should a professional recovery test last?

The appropriate duration depends on the inspection objective and site conditions, but it should be long enough to observe both drawdown and recovery trends rather than relying on a brief flow measurement.

Can two neighboring wells perform differently?

Yes.

Small differences in local geology, fracture systems, and aquifer conditions can produce significant differences in well performance, even between adjacent properties.

Can recovery improve after rainfall?

Heavy rainfall may temporarily increase groundwater recharge in some areas, but a single rainfall event should not be used to evaluate the long-term capability of a private well.

11. Related Knowledge Assets

- KA-002 — Understanding Static Water Level
- KA-003 — Understanding Well Drawdown
- KA-004 — One-Hour Well Performance Testing
- KA-005 — How Recovery Rate Affects Daily Water Supply

12. Revision History

Version	Date	Changes
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2.0	July 2026	First IL-AirTech Knowledge Asset based on real field methodology
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13. References

- Nova Scotia Environment – Private Well Guidance
 - Nova Scotia Groundwater Section
 - CMHC – Private Water Wells
 - InterNACHI® Standards of Practice
 - IL-AirTech field observations
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Figure 1. Field setup used during a continuous well flow test.



Figure 2.

Digital flow meter used to continuously monitor pumping rate throughout the inspection. Continuous measurement provides more reliable results than a single spot measurement.



Figure 3.

Water level measurements recorded during a 60-minute continuous pumping test. The measured drawdown demonstrates how the water level changed while maintaining approximately 4 GPM.

Time	Flow Rate (GPM)	Measured Water Level (Feet)
Pre Test	-----	62.1
Minute 5	3.8	68.89
Minute 15	3.7	78.54
Minute 30	3.6	82.02
Minute 60	4	90.55

Figure 4.

Typical drilled well casing observed during the field inspection. Visual inspection of the well head is an important part of evaluating well protection and accessibility.



Key Takeaways

- ✓ Flow rate alone is insufficient.
- ✓ Recovery matters.
- ✓ Drawdown must be measured.
- ✓ Long-duration testing provides better information.
- ✓ Every well behaves differently.